

PITTSBURNT 2.5

Developer Change Spec

Verified against the uploaded 2.4 implementation - state reset, moving people, UTCI display, balanced Adjust, staged storytelling, and demo-scale UI.

Locked decisions: run the complete 2,409-segment / 167.16-km modelled network; keep exactly Street Tree + Shaded Waiting Shelter; use a balanced-protection default; compare 2026 baseline, future before Adjust, and the same future after Adjust.

What changes in 2.5

- Empty start: no selected time, population, scenario, result, people, thermal tint, hotspot, or budget panel.
- People first: neutral moving agents appear as soon as Who + Time are selected and Moving People is enabled. Crash Test recolours them by computed UTCI and adds exposure halos.
- No false cliff: 38 C UTCI remains the published entry to Very Strong Heat Stress, while the visual ramp and optimizer use continuous heat load above 26 C.
- Balanced Adjust: first protect the highest waiting-exposure unsheltered stop on each Hero corridor, then spend the remainder by deterministic marginal benefit.
- Exact placement: one map icon per purchased unit, actual shade footprints, and post-intervention UTCI arrays returned by the engine.
- Friendly demo UI: larger type, floating rounded cards, explanations on hover/focus, a large centre-bottom Crash button, and a post-crash Adjust modal.
- Full reset: clear React state, Map feature-state, agent data, placements, timers, overlays, and camera state.
- Reproducible build: repair the lockfile and lint error before handoff.

Truth rule

Animate only values on the path to a validated snapshot. The language model may explain a plan, but it never creates costs, effects, placements, or impact numbers.

1. Verified 2.4 audit - what is actually wrong

Priority	2.4 implementation	2.5 correction
P0	Defaults are hard-coded to 2035 / 3 PM / Older adults / \$250K / All.	Make selection fields nullable. Disable Crash until Time + Who + Scenario are complete.
P0	Moving agents require a Crash result and the people reveal stage.	Render neutral agents before Crash after Who + Time. Bind heat colour only after a snapshot exists.
P0	Every agent completes a route in 14 s, regardless of persona speed.	Use real route minutes with one explicit demo playback multiplier. Zoom never changes ground speed.
P0	Reset leaves hotspot feature-state and the last agent source data.	Clear all feature-state keys and empty every dynamic GeoJSON source; cancel timers and animation frames.
P0	Future summary colours the first N time boxes, not the actual crossing hours.	Remove the summary from the main UI; if retained in evidence, store exact crossing_hours.
P0	After-map sun is inferred from an exposure ratio.	Return after_sun, after_utci_c, after_shelter_coverage, and shade geometry directly from the engine.
P0	All 300 cached Across-all plans buy zero shelters.	Replace pure ROI default with the approved balanced-protection policy, clearly labelled as a policy constraint.
P1	A placement is one green dashed segment even when it contains many units.	Return point geometry per unit and render exact tree/shelter counts with staggered icon drops.
P1	SOU / ASS / COM badges are hover-only title attributes.	Make them buttons that open, scroll, focus, and highlight the matching source section.
P1	Panels use 10-12 px text and a 310 px + 350 px dense layout.	Increase demo-critical labels to 16-24 px; move detail to accessible popovers.
P0	npm ci fails because package-lock is out of sync; lint has 1 error.	Regenerate and commit lockfile; fix BudgetField effect; run clean-install CI.

Checks completed: the UTCI engine property suite passes 42/42. A production build succeeds after dependency repair. Full pipeline verification was not executed in this review environment because GeoPandas is not installed; keep it as a release gate.

2. State machine, empty start, and full reset

Start with a context map, not a pre-answered question. Existing protection layers may be toggled independently, but no analytic state is implied until the user makes a selection.

Nullable selection contract

```
interface Selection {
  scenario: ScenarioId | null;
  hour: 8 | 12 | 15 | 18 | null;
  persona: PersonaId | null;
}

initial = {
  selection: { scenario: null, hour: null, persona: null },
  result: null, adapted: null,
  crashStage: 'idle', adaptStage: 'idle',
  layers: { shadow:false, canopy:false, trees:false,
            buildings:false, trips:false, agents:false }
};

canRunCrash = Boolean(scenario && hour && persona);
showBudget = crashStage === 'complete';
```

Interaction flow

State	Visible UI	Primary action
Empty	Context map + three large selectors + optional protection layers	Crash button disabled with “Choose 3 inputs”
Ready	Neutral moving agents if enabled; no red streets or result card	Large red RUN CRASH TEST
Crash complete	Climate / UTCI / Human Exposure result; top hotspot	Crash button morphs to ADJUST
Adjust modal	Budget input + Balanced Protection default + concise method tooltip	BUILD THE PLAN
Adjusted	Exact placements + cooling / protected-wait bloom + three-stage result	REPLAY or RESET

Reset acceptance contract

```
reset():
  cancel crash/adapt controllers and all requestAnimationFrame handles
  abort pending API calls
  set selection and results to null
  clear segment feature-state: scored, utci, severe, sun, hotspot
  empty agents, placements, shade-footprints, trails GeoJSON sources
  close/minimize overlays and remove map popups
  restore context-only camera and layer defaults

assert(dynamicSourceFeatureCount === 0)
assert(noSegmentFeatureStateIsActive)
assert(noTimerOrAnimationIsRunning)
```

3. Layers and moving people

The two tree layers answer different questions. Rename them so the difference is visible without explanation.

Control label	Meaning	Verified 2.4 data	2.5 visual
Existing Tree Inventory	Individual public tree points from the City inventory	2,231 points	Cluster at overview zoom; individual crowns on zoom-in
Existing Canopy Coverage	Overhead foliage raster, including vegetation absent from the point inventory	2010 vintage; 32.3% model area / 36.3% context	Soft green texture below points; show vintage on hover
Moving People	A deterministic visible subset of synthetic routed trips	50 visible agents from the selected cohort	Neutral before Crash; UTCI colour + cumulative halo after Crash

Movement implementation

```
physicalDurationMs = route.minutes * 60_000;
visualDurationMs = physicalDurationMs / PLAYBACK_MULTIPLIER;
u = ((now - agent.startMs) % visualDurationMs) / visualDurationMs;

// Simulation truth remains in metres and seconds.
// Zoom changes pixels-per-metre only; never speed_mps or route.minutes.
// Use one declared multiplier for every persona.
```

Persona speeds - planning assumptions

Persona	Physical speed	Required visual test
Students	1.30 m/s	Same route appears 44% faster than Older adults
Workers	1.20 m/s	Same route appears 33% faster than Older adults
Older adults	0.90 m/s	Never rendered with the common 14-second period
Mobility-constrained	0.80 m/s	Slowest cohort at every zoom level

Agent halo fix: cumulative severe time must integrate each route interval. The current function resets to zero whenever the sampled point falls below 38 C; it is not truly cumulative. Store accumulated dose per agent or precompute a route exposure profile.

4. Climate reveal, UTCI, and Human Exposure

Do not animate humidity, wind, or solar radiation upward. The current scenario method changes air temperature only; RH, wind, and radiation are held constant to isolate the warming delta. Animate temperature and the derived Tmrt / UTCI / heat burden, and label the controls as held constant.

Verified default story - 3 PM, Older adults

State	Air temp	RH / wind / solar	UTCI sun	Heat load	Severe exposure
2026 hot-day baseline	33.05 C / 91.5 F	39% / 2.97 m/s / 765 W/m2	36.40 C	12,238	0 person-min
2035 before Adjust	35.04 C / 95.1 F	held constant	38.85 C	15,525	949.83 person-min
2035 after Balanced Adjust	same future weather	held constant	local values change by shade	15,101.37	838.05 person-min
2050 before Adjust	36.62 C / 97.9 F	held constant	40.82 C	18,162	949.83 person-min

The 2035 and 2050 3 PM severe totals are equal because the binary threshold counts the same sun-exposed minutes once both scenarios cross 38 C. Therefore the main animated burden counter should be continuous Heat Load; show Severe Heat Exposure as the published-class count beside it.

Metric contract

Human Exposure = walking exposure + waiting exposure

```
severe_person_minutes =  
  SUM(person_minutes in sun where UTCI_sun >= 38 C) +  
  SUM(person_minutes in shade where UTCI_shade >= 38 C)
```

cumulative_heat_load =
 SUM(max(UTCI - 26 C, 0) * person_minutes)

Map A: absolute UTCI band on the street
Map B: human hotspot halo / agent dose
Never use exposure rank to recolour the thermal band.

2035 / 3 PM band check

Scope	Active segments	32-38 C Strong	38-46 C Very Strong	Inactive / context
Full modelled network	579	260	319	1,830 of 2,409
Hero corridors	68	24	44	88 of 156

“1 of 4 hours” means one of four sampled times, not one hour of a four-hour period. Remove it from the main screen with Across the day. If retained in Sources, display exact sampled times and actual crossing hours.

5. Balanced Adjust - Tree + Shaded Waiting Shelter

The current optimizer is behaving consistently with its inputs, but not with the desired planning story. Waiting is only 3.4% of the 2035 / 3 PM severe exposure, a tree costs \$1,200, a shelter costs \$15,000, and pure severe-minutes-per-dollar therefore chooses trees in all 300 cached Across-all plans.

Approved objective

Phase A - transparent service floor
For each Hero corridor:
select the eligible unsheltered stop with the highest
computed waiting exposure and place one shaded shelter.

Phase B - deterministic marginal allocation
With the remaining budget, repeatedly choose the feasible unit
with the greatest severe-person-minutes avoided per dollar;
break ties by heat-load avoided, cost, then stable candidate ID.

Label: Balanced Protection policy
Do not label it a pure global optimum.

Verified \$250K default arithmetic and simulation

Plan	Units / spend	Severe exposure	Trade-off
Current pure efficiency	208 trees, 0 shelters / \$249,600	949.83 -> 820.47 (-13.62%)	Best current total reduction; no transit protection
Approved Balanced Protection	170 trees, 3 shelters / \$249,000; \$1,000 left	949.83 -> 838.05 (-11.77%)	Protects one highest-wait site per Hero corridor

The balanced result above is recomputed from dataset version 6a0955d7 under the current prototype costs. If the cost catalog or exposure model changes, regenerate the snapshot and never keep these values as UI constants.

Service-floor sites for the verified default

Hero corridor	Selected segment ID	One-shelter waiting severe minutes avoided
Forbes Avenue	mckee-place-0431-13	0.4930
Fifth Avenue	fifth-avenue-0556-01	1.0859
South Craig Street	south-craig-street-1183-00	0.4778

Evidence limitation: stop service frequency is not stop ridership. The model assumes 20% of walking trips end in a transit wait (10-30% sensitivity range). Present the shelter service floor as a planning policy choice, not as an unconstrained AI discovery.

6. Adjust payload and exact map choreography

The UI must not reverse-engineer cooling from an exposure score. The engine owns all post-intervention environmental arrays and exact point placements.

Required response fields

```
interface UnitPlacement {
  unitId: string;
  kind: 'tree' | 'shaded_shelter';
  segmentId: string;
  stopId?: string;
  lon: number;
  lat: number;
  costUsd: number;
  order: number;
}

interface AdaptResult {
  placements: UnitPlacement[]; // one record per purchased unit
  afterSun: number[];
  afterUtcic: number[];
  afterShelterCoverage: number[];
  shadeFootprints: FeatureCollection;
  before: ExposureSnapshot;
  after: ExposureSnapshot;
  policy: 'balanced_protection';
  inputHash: string;
}

assert(placements.length === treeCount + shelterCount);
```

Ten-second Adjust animation

Time	Beat	Map / panel behaviour
0.0-1.0 s	Lock	Freeze budget, policy, scenario, agents, and input hash.
1.0-5.0 s	Place	Drop bundled emoji-style tree and shelter sprites in optimizer order; counts and spend tick to exact totals.
5.0-7.0 s	Protect	Tree shade blooms along footways; shelter roof bloom stays at WAIT_NODE. Do not cool the whole road for a shelter.
7.0-9.0 s	Re-test	Replay identical agents and routes; agent colour / halo follows the returned after arrays.
9.0-10.0 s	Land	Show 2026 baseline, future before, future after; place Replay and Reset.

Use bundled SVG or PNG emoji-style sprites instead of operating-system emoji glyphs. This preserves the requested friendly look while keeping symbol size and appearance deterministic on the demo machine.

Tree units on one segment must have distinct points sampled along the eligible footway. Shelters use the exact validated stop coordinate. At overview zoom cluster icons only after the drop animation; the result panel always shows the full exact count.

7. Demo-first UI redesign

Use the reference images as layout principles: map-first composition, soft floating surfaces, generous spacing, large controls, and data-bound cooling blooms. Do not copy their content or decorative heat spots.

Area	2.5 requirement
Left setup card	Floating rounded card, 360-400 px; only Time, Existing Protection, Who, Future Scenario. Section help appears on hover/focus.
Primary CTA	Centre-bottom circular red button, at least 96 px, strong label and warning pulse. Disabled until three required inputs exist.
Post-crash Adjust	Do not show budget initially. Morph the Crash CTA to Adjust; open budget and policy in a focused overlay.
Right result card	Fixed floating card within viewport. Large Air Temperature, UTCI, Heat Load, Severe Exposure, then top hotspot. Detail lives in popovers.
Overlays	Every overlay has Minimize, Restore, and Close. Keep map controls reachable. Panels do not move when the map pans or zooms.
Responsive gate	Nothing clips at 1366x768 or 1920x1080. Use max-height: calc(100dvh - 32px) and internal scrolling only when expanded.
Provenance badges	SOU / COM / ASS / MIS are buttons. Click opens Sources, scrolls to data-key, focuses it, and applies a 1.5 s highlight.

Remove from the default demo surface

- Across the day and the confusing “N/4 hours” summary.
- Top-five hotspot list - show the top one and a View all drawer.
- Planning weights, raw synthetic trip counts, segment counts, and model disclaimers - move to Sources.
- Long descriptions under every selector, live-engine status text, and intervention unit-cost detail before Adjust.
- Dense multi-row legend - keep one compact UTCI gradient with 26 / 32 / 38 / 46 markers.

Crash choreography - 7.5 seconds

0.0-1.5 Climate: 91.5 F -> selected future temperature
RH / wind / solar remain visibly LOCKED
1.5-3.0 Thermal: Tmrt and UTCI rise; gradient crosses bands
3.0-5.5 People: neutral agents recolour; exposure dose accumulates
5.5-7.5 Hotspot: top human-exposure location pulses and result lands

Counters finish on exact snapshot values even if frames drop or Skip is used.

8. Release gates and developer handoff

P0 - must pass before the next demo

Gate	Pass condition
Empty start	No selected pills, no result, no people, no thermal/hotspot layer, no budget panel.
People toggle	After Who + Time, neutral agents appear without running Crash.
Speed truth	Route duration derives from speed_mps; same cohort speed is invariant under zoom.
2035 verification	At 3 PM / Older adults: 579 active = 260 Strong + 319 Very Strong; no percentile recolouring.
Reset	After Crash or Adjust, zero dynamic features and zero active feature-state remain.
Adjust truth	Exact icon count equals payload count; shelter changes waiting coverage, not street sun.
Balanced default	\$250K / dataset 6a0955d7 returns 170 trees + 3 shelters only while current cost/model inputs match.
Three-state result	2026 baseline -> selected future before -> same future after; never compare 2026 directly to adjusted future as intervention effect.
Build	Fresh npm ci, npm run lint, npm run build, 42 engine tests, pipeline verify, and deterministic replay all pass.

P1 - polish after P0

- Viewport-safe minimize/restore behaviour and keyboard-accessible tooltips.
- Icon clustering after placement; exact unit detail on zoom or click.
- Smooth camera easing to the top hotspot and back to the full network.
- Reduced-motion path that keeps the same ordering and exact final values.

Known data limitations that must stay visible in Sources

Input	Status / limitation
Tree \$1,200; shelter \$15,000	Low-confidence planning assumptions, not Pittsburgh procurement quotes.
20% transit-trip share	Low-confidence assumption; expose 10-30% sensitivity.
Walking trips / persona mix	Synthetic fixed-seed samples, not calibrated pedestrian counts or Census shares.
Canopy	2010 raster; may omit recent planting and removals.
Building heights	59.5% measured or sourced; remainder modelled with record-level confidence.
Future weather	CMIP6-LOCA2 temperature delta applied to observed hot-day profile; RH, wind, and radiation held constant.
Tree feasibility / growth	Current capacity is segment-length based and mature shade is immediate; utilities, pits, survival, and Year-1 growth are not yet modelled.
Shelter demand	Service frequency proxy plus assumed trip share; no stop-level boarding calibration.

Final product sentence

Pittsburnt does not promise a medically safe street. It shows where a reproducible future heat scenario overlaps with human walking and waiting, then makes the policy trade-off behind a budgeted shade plan visible.